Calculus

Complete Study Notes

Contents

1. Limits

1.1 The Idea of a Limit

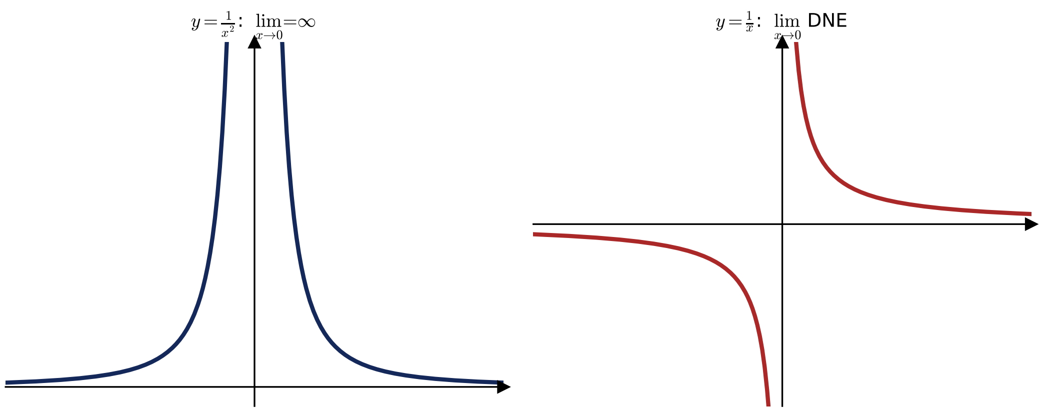
A **limit** is the value a function approaches as the input variable gets nearer and nearer to a particular value.

1.2 Estimating Limits from Graphs

Consider $y = \dfrac{1}{x^2}$ versus $y = \dfrac{1}{x}$ near 0:

$$\lim_{x\to 0} \frac{1}{x^2} = \infty, \qquad \lim_{x\to 0} \frac{1}{x} \text{ does not exist (DNE).}$$

For $\frac{1}{x^2}$ the sign of x does not matter (always positive), so both sides go to $+\infty$. For $\frac{1}{x}$, the sign of x matters: the left and right limits disagree.



Near $x = 0$, $\frac{1}{x^2}$ blows up to $+\infty$ on both sides, while $\frac{1}{x}$ splits.

1.3 Limit Laws (Algebraic Properties)

If $\lim_{x\to a} f(x) = L_1$ and $\lim_{x\to a} g(x) = L_2$, then

$$\begin{aligned} \lim_{x\to a} k\, f(x) &= k\, L_1, \\ \lim_{x\to a} \big[f(x) + g(x)\big] &= L_1 + L_2, \\ \lim_{x\to a} \big[f(x)\cdot g(x)\big] &= L_1 \cdot L_2. \end{aligned}$$

1.4 Algebraic Manipulation

Factor the function first when direct substitution fails.

Rule 1. If k and n are constants with $n > 0$, then $\lim_{x\to\infty} \frac{k}{x^n} = 0$.

Rule 2. When an expression is one polynomial divided by another, divide *each term* of numerator and denominator by the highest power of x.

Example.

$$\lim_{x\to\infty} \frac{3x+5}{7x-2} = \lim_{x\to\infty} \frac{3+\frac{5}{x}}{7-\frac{2}{x}} = \frac{3}{7}.$$

1.5 One-Sided Limits and DNE

$$\lim_{x\to a^-} f(x) \quad \text{and} \quad \lim_{x\to a^+} f(x).$$

If the left-hand limit $\neq$ right-hand limit, the limit **DNE**. Useful infinite-limit facts:

$$\lim_{x\to 0^+} \frac{k}{x} = \infty, \quad \lim_{x\to 0^-} \frac{k}{x} = -\infty, \quad \lim_{x\to 0} \frac{k}{x} = \text{DNE}, \quad \lim_{x\to 0} \frac{k}{x^2} = \infty.$$

1.6 The Squeeze Theorem

If $g(x) \le f(x) \le h(x)$ for all x in an interval containing a, and $\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L$, then $\lim_{x\to a} f(x) = L$.

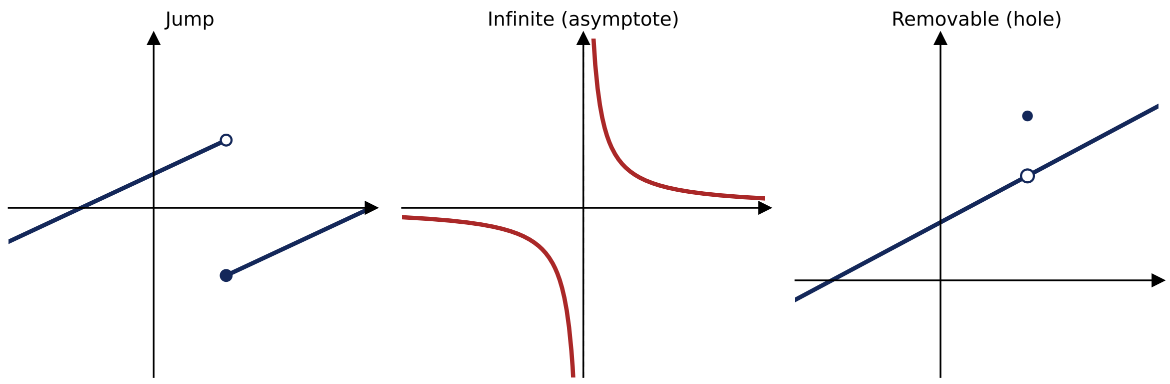
1.7 Special Trigonometric and Exponential Limits

$$\lim_{x\to 0} \frac{\sin x}{x} = 1, \qquad \lim_{x\to 0} \frac{\cos x - 1}{x} = 0, \qquad \lim_{x\to 0} \frac{\sin ax}{\sin bx} = \frac{a}{b},$$
$$\lim_{x\to 0} \frac{\sin ax}{x} = a, \qquad \lim_{n\to\infty} \left(1+\frac{1}{n}\right)^n = e.$$

2. Continuity

2.1 Types of Discontinuity

- **Jump discontinuity:** the curve “breaks” at a point and starts elsewhere. Both one-sided limits exist but do not match.
- **Infinite (essential) discontinuity:** appears where there is a vertical asymptote.
- **Removable discontinuity:** an otherwise continuous curve has a single “hole.”



The three discontinuity types. Open circle = value not attained; filled circle = actual value.

2.2 Continuity at a Point

$f(x)$ is continuous at $x = c$ when all three hold:

1. $f(c)$ exists,
2. $\lim_{x \to c} f(x)$ exists,
3. $\lim_{x \to c} f(x) = f(c)$.

2.3 Asymptotes of Rational Functions

For $y(x) = \dfrac{a_n x^n}{a_m x^m}$:

- $n > m$: no horizontal asymptote;
- $n = m$: horizontal asymptote $y = \dfrac{a_n}{a_m}$;
- $n < m$: horizontal asymptote $y = 0$.

Vertical asymptote: set the denominator $= 0$.

2.4 Intermediate Value Theorem (IVT)

If f is continuous on $[a, b]$ and C is any number between $f(a)$ and $f(b)$, then there exists at least one x in $[a, b]$ with $f(x) = C$.

3. The Derivative: Definition and Basic Rules

3.1 Rates of Change

Average rate of change (over an interval):

$$\frac{f(a+h) - f(a)}{h} \qquad \text{or} \qquad \frac{f(x) - f(a)}{x - a}.$$

Instantaneous rate of change (at an instant):

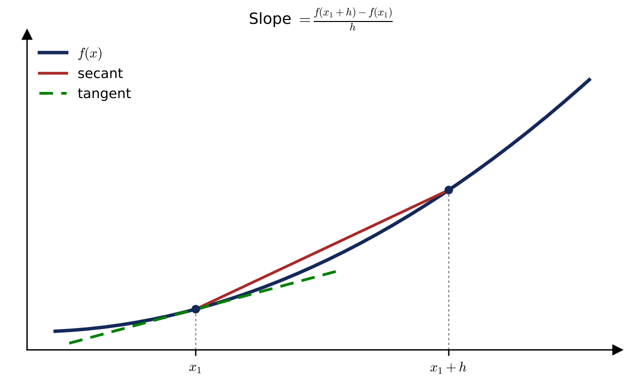
$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \qquad \text{or} \qquad \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

3.2 Slope of a Curve and the Difference Quotient

The tangent line is the limit of secant lines. The slope of a secant line through two points is the **difference quotient**:

$$\frac{f(x_1+h)-f(x_1)}{(x_1+h)-x_1} = \frac{f(x_1+h)-f(x_1)}{h}.$$

The closer the two points, the more accurate the approximation; the tangent line is the most accurate approximation at one point.



As $h \to 0$ the secant line rotates into the tangent line.

3.3 Definition of the Derivative

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}.$$

3.4 Basic Differentiation Rules

$$\begin{aligned}
\text{Power rule:}\quad & y = x^n \Rightarrow \frac{dy}{dx} = nx^{n-1},\\
\text{Constant rule:}\quad & f(x) = k \Rightarrow f'(x) = 0, \quad \tfrac{d}{dx}(k) = 0,\\
\text{Constant multiple:}\quad & \big[k\,f(x)\big]' = k\,\frac{d}{dx}\big[f(x)\big],\\
\text{Sum rule:}\quad & y = ax^n + bx^m \Rightarrow \frac{dy}{dx} = a\,nx^{n-1} + b\,mx^{m-1}.
\end{aligned}$$

The derivative of a constant is always 0.

3.5 Derivatives of Common Functions

$$\frac{d}{dx}\sin x = \cos x, \qquad \frac{d}{dx}\cos x = -\sin x, \qquad \frac{d}{dx}e^x = e^x, \qquad \frac{d}{dx}\ln x = \frac{1}{x}.$$

4. Differentiation Techniques

4.1 The Chain Rule

$$\frac{d}{dx}\left[f(g(x))\right] = f'(g(x))\,g'(x).$$

If $y = y(v)$ and $v = v(x)$, then $\dfrac{dy}{dx} = \dfrac{dy}{dv} \cdot \dfrac{dv}{dx}$. Examples:

$$y = \ln u \Rightarrow \frac{dy}{dx} = \frac{1}{u}\frac{du}{dx}, \qquad y = e^u \Rightarrow \frac{dy}{dx} = e^u \frac{du}{dx}.$$

4.2 Logarithmic and Exponential Derivatives

$$y = \log_a x \Rightarrow \frac{dy}{dx} = \frac{1}{x \ln a}, \qquad y = \log_a u \Rightarrow \frac{dy}{dx} = \frac{1}{u \ln a}\frac{du}{dx}.$$

Since $a^x = e^{x \ln a}$:

$$y = a^x \Rightarrow \frac{dy}{dx} = a^x \ln a.$$

4.3 Product and Quotient Rules

$$f = uv \Rightarrow f' = u\frac{dv}{dx} + v\frac{du}{dx}, \qquad f = \frac{u}{v} \Rightarrow f' = \frac{v\,\frac{du}{dx} - u\,\frac{dv}{dx}}{v^2}.$$

4.4 Derivatives of the Other Trig Functions

Derived from quotient rule and Pythagorean identities:

$$\begin{aligned}
\frac{d}{dx}\tan x &= \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \sec^2 x,\\
\frac{d}{dx}\cot x &= \frac{\sin x(-\sin x) - \cos x \cos x}{\sin^2 x} = \frac{-1}{\sin^2 x} = -\csc^2 x,\\
\frac{d}{dx}\sec x &= \sec x \tan x,\\
\frac{d}{dx}\csc x &= -\csc x \cot x.
\end{aligned}$$

4.5 Differentiability and Continuity

$f'(c) = \lim\limits_{x \to c} \dfrac{f(x) - f(c)}{x - c}$.

1. If f is differentiable at $x = c$, then f is continuous at $x = c$.
2. If f is not continuous at $x = c$, then f is not differentiable at $x = c$.

No derivative exists at: (1) a vertical tangent, (2) a discontinuity, (3) a “sharp” corner.

4.6 Composite, Implicit, and Inverse Functions

Implicit differentiation: when an equation mixes x and y, differentiate term by term (using the chain rule on y), then solve for $\frac{dy}{dx}$.

$$\text{e.g. } y^2 + y = 3x^6 - 7x \Rightarrow 2y\frac{dy}{dx} + \frac{dy}{dx} = 18x^5 - 7 \Rightarrow \frac{dy}{dx} = \frac{18x^5 - 7}{2y+1}.$$

Inverse function derivative:

$$g'(x) = \frac{1}{f'(g(x))}, \qquad \left[f^{-1}(x)\right]' = \frac{1}{f'(f^{-1}(x))}.$$

4.7 Derivatives of Inverse Trig Functions

$$\frac{d}{dx}\sin^{-1} u = \frac{1}{\sqrt{1-u^2}}\frac{du}{dx}, \qquad \frac{d}{dx}\cos^{-1} u = \frac{-1}{\sqrt{1-u^2}}\frac{du}{dx} \quad (-1 < u < 1),$$
$$\frac{d}{dx}\tan^{-1} u = \frac{1}{1+u^2}\frac{du}{dx}, \qquad \frac{d}{dx}\cot^{-1} u = \frac{-1}{1+u^2}\frac{du}{dx},$$
$$\frac{d}{dx}\sec^{-1} u = \frac{1}{|u|\sqrt{u^2-1}}\frac{du}{dx}, \qquad \frac{d}{dx}\csc^{-1} u = \frac{-1}{|u|\sqrt{u^2-1}}\frac{du}{dx} \quad (|u| > 1).$$

Proof sketch (for $\sin^{-1}$): from $\sin y = x$, $\cos y \frac{dy}{dx} = 1$, so $\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}} = \frac{1}{\sqrt{1-x^2}}$.

4.8 Higher-Order Derivatives

Differentiate repeatedly, e.g.

$$\frac{dy}{dx}x^6 = 6x^5, \qquad \frac{d^2y}{dx^2} = 30x^4.$$

5. Applications of Differentiation

5.1 Position, Velocity, Acceleration

$$v(t) = x'(t), \qquad a(t) = x''(t) = v'(t).$$

To find the distance a car drove in a time interval: (1) find when the velocity changes sign, (2) account for direction. Identify whether the velocity is positive or negative.

5.2 Related Rates

Situations where the rate of change of one quantity is related to the rate of change of another.

Example. A circle's area grows at 16π in^2/s. At what rate is the radius expanding when $r = 4$ in?

$$\frac{dA}{dt} = 2\pi r\frac{dr}{dt} \Rightarrow 16\pi = 2\pi(4)\frac{dr}{dt} \Rightarrow 16\pi = 8\pi\frac{dr}{dt} \Rightarrow \frac{dr}{dt} = 2 \text{ in/s}.$$

5.3 Tangent and Normal Lines

Tangent/normal line through (x_1, y_1) with slope m:

$$y - y_1 = m(x - x_1).$$

A **normal line** is perpendicular to the tangent line at the same point (slope $-\frac{1}{m}$).

5.4 Linearization, Differentials, and Newton's Method

A differential is a very small change in a quantity. The linear approximation:

$$f'(x) = \frac{f(x + \Delta x) - f(x)}{\Delta x} \ \Rightarrow\ f(x + \Delta x) \approx f(x) + f'(x)\,\Delta x.$$

Example. Approximate $\sqrt{9.01}$ with $x = 9,\ \Delta x = 0.01,\ f(x) = \sqrt{x},\ f'(x) = \frac{1}{2\sqrt{x}}$:

$$f(9.01) \approx \sqrt{9} + \frac{1}{2\sqrt{9}}(0.01) \approx 3.00167.$$

Differential form: $dy = f'(x)\,dx$ estimates errors / effects of small changes.
Newton's Method: to solve $f(x) = 0$, start at $x_1 = a$ and iterate

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}, \qquad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

5.5 L'Hôpital's Rule

A way to find limits of indeterminate forms $\frac{0}{0}$ or $\frac{\infty}{\infty}$. If $f(c) = g(c) = 0$ (or ∞), f', g' exist, and $g'(c) \neq 0$, then

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}.$$

Other indeterminate forms:

- $0 \cdot \infty$: rewrite $f(x)g(x)$ as $\dfrac{f(x)}{1/g(x)}$, then apply the rule.
- $\infty - \infty$: convert to a single fraction, then apply the rule.
- $\infty^0,\ 0^0,\ 1^\infty$: let $y = f(x)^{g(x)}$, take $\ln y = g(x) \ln f(x)$, then apply the rule.

5.6 Mean Value, Rolle's, and Extreme Value Theorems

Mean Value Theorem (MVT). If f is continuous on $[a, b]$ and differentiable on (a, b), there is at least one c in (a, b) with

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Example: $f(x) = x^2$ on $[1, 3]$, $f'(c) = \dfrac{9 - 1}{3 - 1} = 4.$

Rolle's Theorem (special case of MVT). If additionally $f(a) = f(b)$, there is a c in (a, b) with $f'(c) = 0$.

Extreme Value Theorem (EVT). If f is continuous on $[a, b]$, then f attains at least one maximum and one minimum value on $[a, b]$.

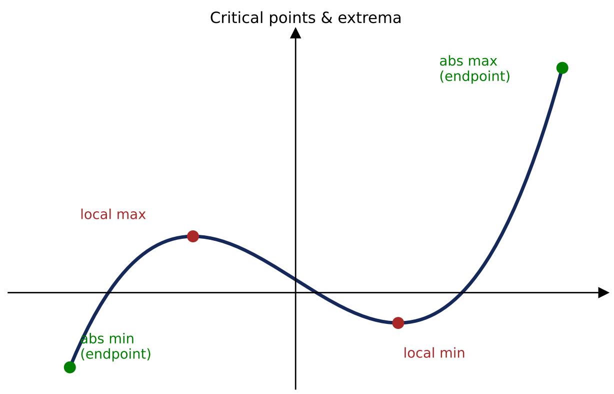
Tip: when unsure what to do, take the derivative of the equation and set it equal to zero.

5.7 Extrema and Critical Points

- **Absolute extremum:** no other value of f is higher (max) / lower (min).
- **Local (relative) extremum:** the highest / lowest value in its neighborhood.

Use the derivative to determine the tendency:

$$f'(a) > 0: \text{ increasing,} \qquad f'(a) < 0: \text{ decreasing,} \qquad f'(a) = 0: \text{ critical point.}$$



Local extrema occur at interior critical points; absolute extrema may occur at endpoints.

5.8 First Derivative Test

If f is continuous, $f'(a) = 0$, and $f'(x) > 0$ to the left of $x = a$ while $f'(x) < 0$ to the right, then f has a relative **maximum** at a (reverse the signs for a minimum).

5.9 Candidates Test

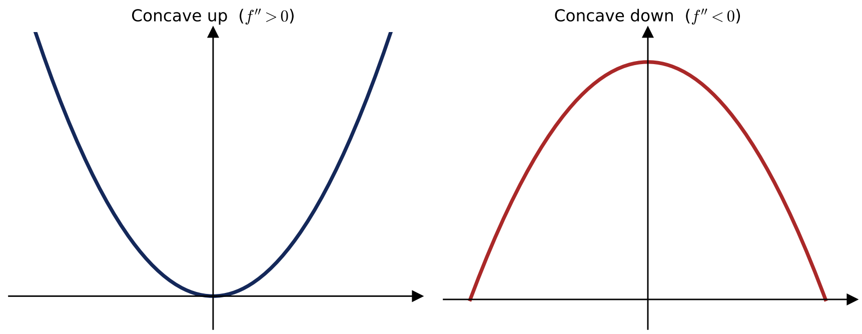
Evaluate f at all critical points *and* the endpoints. The largest value is the absolute maximum; the smallest is the absolute minimum.

5.10 Concavity and the Second Derivative Test

- First derivative → whether a function is increasing / decreasing.
- Second derivative → concavity / curvature.
- $f'' > 0$: concave up; $f'' < 0$: concave down; $f'' = 0$: possible inflection point.

Second Derivative Test. If $f'(a) = 0$:

$$f''(a) > 0 \Rightarrow \text{minimum at } (a, f(a)), \qquad f''(a) < 0 \Rightarrow \text{maximum at } (a, f(a)).$$



Concave up holds water ($f'' > 0$); concave down spills it ($f'' < 0$).

5.11 Curve Sketching

1. Test the function: $f(x) = 0 \Rightarrow x$-intercepts (and find the y-intercept).
2. $f'(x) = 0 \Rightarrow$ critical points.
3. $f''(x) = 0 \Rightarrow$ inflection points / concavity (curve up or down).
4. Test end behavior.

Then plug the original x-values into the original equation to recover the points.

5.12 Finding a Cusp

If the function approaches ∞ from one side of a point and $-\infty$ from the other, and the function is continuous there, the curve has a *cusp*. To find one, look where the first derivative is undefined as well as where it is zero.

5.13 Optimization

Use $f'(x)$ to find critical points, then use $f''(x)$ to classify each:

$$f''(c) < 0 \Rightarrow \text{maximum}, \qquad f''(c) > 0 \Rightarrow \text{minimum}.$$

6. Integration and Accumulation of Change

6.1 The Definite Integral

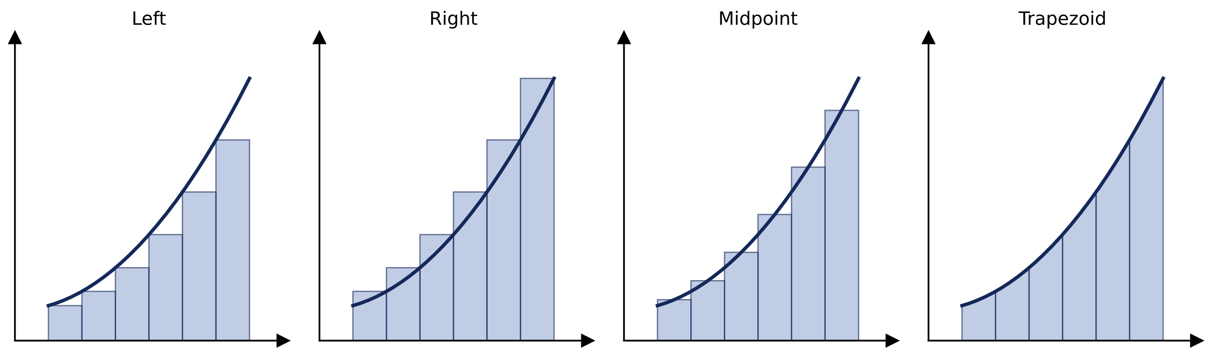
A definite integral gives the area of the region between a function and the x-axis.

6.2 Approximating Area with Riemann Sums

Let a be the left endpoint, b the right endpoint, n the number of rectangles, and width $\Delta x = \dfrac{b-a}{n}$; y values are the heights.

$$\begin{aligned}
\textbf{Left endpoint:}\quad & S = \frac{b-a}{n}\big[y_0 + y_1 + y_2 + \cdots + y_{n-1}\big],\\
\textbf{Right endpoint:}\quad & S = \frac{b-a}{n}\big[y_1 + y_2 + y_3 + \cdots + y_n\big],\\
\textbf{Midpoint:}\quad & S = \frac{b-a}{n}\big[y_{1/2} + y_{3/2} + y_{5/2} + \cdots + y_{(2n-1)/2}\big],\\
\textbf{Trapezoid:}\quad & S = \frac{1}{2}\,\frac{b-a}{n}\big[y_0 + 2y_1 + 2y_2 + \cdots + 2y_{n-1} + y_n\big].
\end{aligned}$$

Letting $n \to \infty$ creates infinitely many, infinitesimally thin rectangles — the exact area.



Four ways to approximate area: left, right, midpoint rectangles, and trapezoids.

6.3 Riemann Sum and Definite-Integral Notation

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \Delta x \sum_{i=1}^{n} f(a + i\,\Delta x), \qquad \Delta x = \frac{b-a}{n}.$$

7. The Fundamental Theorem of Calculus

7.1 FTC and Accumulation Functions

A definite integral evaluates the antiderivative and finds the difference between the antiderivative at two points. (Integral = antiderivative.) If f is continuous on $[a,b]$ and $F(x) = \int_a^x f(t)\,dt$, then

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x)\cdot\frac{dx}{dx} = f(x).$$

If the upper limit is a function of x rather than just x, multiply by the derivative of that term:

$$\frac{d}{dx}\int_{g(x)}^{h(x)} f(t)\,dt = f(h(x))\,h'(x) - f(g(x))\,g'(x).$$

Example.

$$\frac{d}{dx}\int_2^{x^2}(1+t^3)\,dt = \big[1+(x^2)^3\big](2x).$$

The accumulation function $F(x)=\displaystyle\int_0^x f(t)\,dt$ calculates the area under the curve.

7.2 FTC (Evaluation Form)

$$\int_a^b f(x)\,dx = F(b)-F(a), \qquad F(x) \text{ an antiderivative of } f(x).$$

7.3 Second Fundamental Theorem

$$\frac{d}{dx}\int_c^u f(t)\,dt = f(u)\,\frac{du}{dx}.$$

7.4 Properties of Definite Integrals

Let $\displaystyle\int_a^b f(x)\,dx = M$ and $\displaystyle\int_b^c f(x)\,dx = N$. Then

$$\int_a^b k\,f(x)\,dx = kM, \qquad \int_a^b f\,dx + \int_b^c f\,dx = M+N = \int_a^c f(x)\,dx,$$
$$\int_b^a f(x)\,dx = -M, \qquad \int_a^b \big[f\pm g\big]\,dx = \int_a^b f\,dx \pm \int_a^b g\,dx.$$

8. Techniques of Integration

8.1 Basic Rules

$$\int x^n\,dx = \frac{x^{n+1}}{n+1}+C\ (n\neq -1), \quad \int k\,dx = kx+C, \quad \int x^{-1}\,dx = \ln|x|+C,$$
$$\int e^x\,dx = e^x+C, \qquad \int \ln x\,dx = x\ln x - x + C.$$

8.2 Integrals of Trig Functions

$$\int \sin ax\,dx = -\frac{\cos ax}{a}+C, \qquad \int \cos ax\,dx = \frac{\sin ax}{a}+C,$$
$$\int \sec^2 ax\,dx = \frac{\tan ax}{a}+C, \qquad \int \csc^2 ax\,dx = -\frac{\cot ax}{a}+C,$$
$$\int \sec ax\tan ax\,dx = \frac{\sec ax}{a}+C, \qquad \int \csc ax\cot ax\,dx = -\frac{\csc ax}{a}+C.$$

8.3 Integration by Substitution (u-substitution)

$$\int u^n\,du = \frac{u^{n+1}}{n+1} + C.$$

Steps: (1) choose u, (2) compute du, (3) substitute into the integral, (4) integrate.

Example.

$$\int 10x(5x^2-3)^6\,dx, \quad u = 5x^2-3,\ du = 10x\,dx \ \Rightarrow\ \int u^6\,du = \frac{u^7}{7} + C = \frac{(5x^2-3)^7}{7} + C.$$

8.4 More Trigonometric Integrals

$$\int \tan x\,dx = -\ln|\cos x| + C, \qquad \int \cot x\,dx = \ln|\sin x| + C,$$
$$\int \sec x\,dx = \ln|\sec x + \tan x| + C, \qquad \int \csc x\,dx = -\ln|\csc x + \cot x| + C.$$

Derivation of $\int \sec x\,dx$: multiply by $\dfrac{\sec x + \tan x}{\sec x + \tan x}$, giving $\displaystyle\int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x}\,dx$, then u-substitute $u = \sec x + \tan x$.

8.5 Integrating e^x and a^x

$$\int e^u\,du = e^u + C, \qquad \int e^{kx}\,dx = \frac{1}{k}e^{kx} + C, \qquad \int a^u\,du = \frac{1}{\ln a}\,a^u + C.$$

8.6 Powers of Sine and Cosine

$$\int \sin^n x \cos x\,dx = \frac{\sin^{n+1} x}{n+1} + C, \qquad \int \cos^n x \sin x\,dx = -\frac{\cos^{n+1} x}{n+1} + C.$$

8.7 Integration by Parts

$$\int u\,dv = uv - \int v\,du.$$

8.8 Integrals Yielding Inverse Trig Functions

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C, \qquad \int \frac{dx}{1+x^2} = \tan^{-1} x + C, \qquad \int \frac{dx}{x\sqrt{x^2-1}} = \sec^{-1} x + C.$$

Example.

$$\int \frac{x\,dx}{\sqrt{1-x^4}}, \quad u = x^2,\ du = 2x\,dx \ \Rightarrow\ \frac{1}{2}\int \frac{du}{\sqrt{1-u^2}} = \frac{1}{2}\sin^{-1} x^2 + C.$$

8.9 Long Division and Completing the Square

If the numerator's degree $\geq$ the denominator's degree, divide first. **Example.**

$$\int \frac{4x^2 - 14x + 9}{x - 3}\,dx = \int \left(4x - 2 + \frac{3}{x-3}\right) dx = 2x^2 - 2x + 3\ln|x-3| + C.$$

8.10 Partial Fractions

$$\frac{A}{a-b} + \frac{B}{b-c} = \frac{1}{(a-b)(b-c)}.$$

Example. For $\displaystyle\int \frac{19x - 16}{(2x-3)(x+1)}\,dx$, write

$$\frac{A}{2x-3} + \frac{B}{x+1} = \frac{19x-16}{(2x-3)(x+1)} \Rightarrow A(x+1) + B(2x-3) = 19x - 16.$$

Solving ($x = -1 \Rightarrow B = 7$; $x = \frac{3}{2} \Rightarrow A = 5$):

$$\int \frac{5}{2x-3}\,dx + \int \frac{7}{x+1}\,dx = \frac{5}{2}\ln|2x-3| + 7\ln|x+1| + C.$$

9. Differential Equations

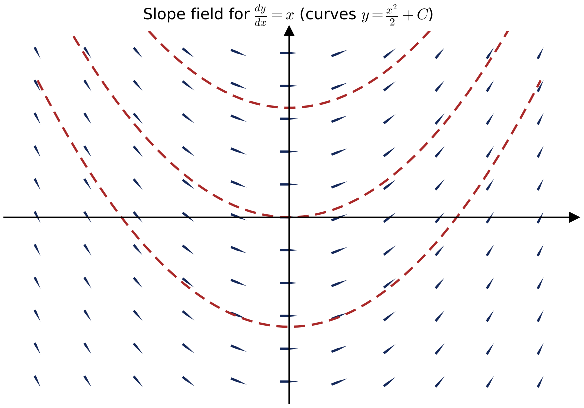
9.1 General vs. Particular Solutions

- **General solution:** uses a constant to represent all possible solutions.
- **Particular solution:** substitutes the constant C with a specific number.

Example. For $\frac{dy}{dt} = 6t^2 - 2t + 7$, verify $y = 2t^3 - t^2 + 7t + 5$: $y' = 6t^2 - 2t + 7$ ✓, so y is a particular solution.

9.2 Slope Fields

A slope field is a graphical representation of the slope of a function at various points in the plane. *Example:* given $\frac{dy}{dx} = x$, the slope at each point equals its x-coordinate (e.g. at $x = 1$, slope $= 1$; at $x = -1$, slope $= -1$).



Slope field for $\frac{dy}{dx} = x$; the dashed solution curves are $y = \frac{x^2}{2} + C$.

9.3 Euler's Method

Approximate solutions step by step:

$$x_n = x_{n-1} + h, \qquad y_n = y_{n-1} + h \cdot f(x_{n-1}, y_{n-1}), \qquad n = 1, 2, 3, \ldots$$

9.4 Separation of Variables

Move all terms with one variable to one side and all terms with the other variable to the other side, then integrate both sides. **Example.**

$$\frac{dy}{dx} = \frac{5x^2}{6y^2} \Rightarrow 6y^2\,dy = 5x^2\,dx \Rightarrow 2y^3 = \frac{5}{3}x^3 + C \Rightarrow y = \left(\frac{5}{6}x^3 + \frac{C}{2}\right)^{1/3}.$$

9.5 Particular Solutions with Initial Conditions

Example. If $\frac{dy}{dx} = 3x^2 y$ with $y(0) = 2$:

$$\frac{1}{y}\,dy = 3x^2\,dx \Rightarrow \ln|y| = x^3 + C \Rightarrow y = Ce^{x^3}.$$

Apply $y(0) = 2$: $2 = Ce^0 \Rightarrow C = 2$, so $y = 2e^{x^3}$.

9.6 Exponential and Logistic Models

Exponential: $\dfrac{dB}{B} = k\,dt \Rightarrow \ln B = kt + C \Rightarrow B = Ce^{kt}$. *Example:* $B(0) = 100,\ B(2) = 160$ gives $C = 100,\ 160 = 100e^{2k} \Rightarrow k = \frac{1}{2}\ln\frac{8}{5}$, so $B = 100\left(\frac{8}{5}\right)^{t/2}$.

Logistic:

$$\frac{dP}{dt} = rP(K - P), \qquad P = \frac{P_0 K e^{rt}}{(K - P_0) + P_0 e^{rt}},$$

where r is the growth rate, P the population, P_0 the initial population, and K the carrying capacity.

10. Applications of Integration

10.1 Mean Value Theorem for Integrals (MVTI)

If f is continuous on $[a, b]$, then for some c in $[a, b]$:

$$\int_a^b f(x)\,dx = f(c)(b - a).$$

The area under the curve equals the value of the function at some c times the interval length. The **average value** of f on $[a, b]$:

$$f(c) = \frac{1}{b - a}\int_a^b f(x)\,dx.$$

Example. Average value of $f(x) = 4x\cos x^2$ on $\left[0, \sqrt{\frac{\pi}{3}}\,\right]$ is $\dfrac{1}{\sqrt{\pi/3}} \displaystyle\int_0^{\sqrt{\pi/3}} 4x\cos x^2\, dx$ (use $u = x^2$).

10.2 Position, Velocity, Acceleration via Integrals

$$p'(t) = v(t), \quad p''(t) = a(t), \qquad \int a(t)\,dt = v(t) + v_0, \qquad \int v(t)\,dt = s(t) + s_0.$$

10.3 Area Between Curves

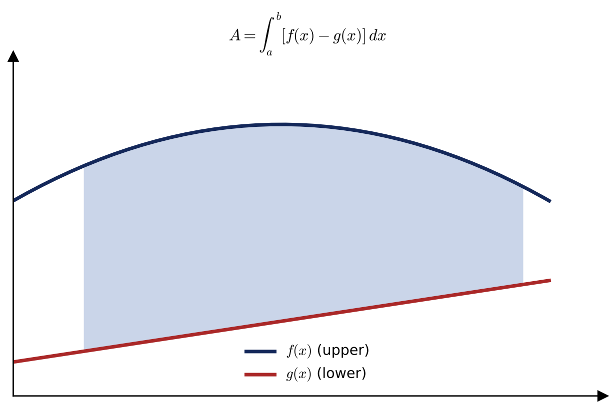
In terms of x: if the region is bounded above by $f(x)$ and below by $g(x)$ on $[a, b]$:

$$A = \int_a^b \left[f(x) - g(x)\right] dx.$$

In terms of y: if bounded on the right by $f(y)$ and on the left by $g(y)$ on $[c, d]$:

$$A = \int_c^d \left[f(y) - g(y)\right] dy.$$

If the curves intersect at more than two points, divide the region into pieces.



The shaded region is the integral of (upper curve − lower curve).

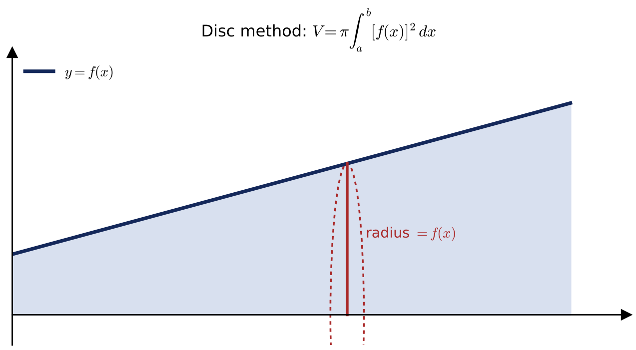
10.4 Volumes by Cross Sections

1. Find the side of the cross section in terms of y (or x).
2. Plug the side into the area formula for the cross section.
3. Integrate the area from one endpoint to the other.

10.5 Volumes of Revolution

Disc method (revolve about the x- or y-axis): each disc has radius $f(x)$, area $\pi[f(x)]^2$:

$$V = \pi \int_a^b \left[f(x)\right]^2 dx.$$



Revolving the region about the x-axis sweeps out discs of radius $f(x)$.

Washer method (solid that does not touch the axis): bounded above by $f(x)$ and below by $g(x)$:

$$V = \pi \int_a^b \left(\left[f(x)\right]^2 - \left[g(x)\right]^2 \right) dx.$$

10.6 Arc Length

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx.$$

11. Parametric, Vector-Valued, and Polar Functions

11.1 Differentiating Parametric Equations

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

Example. $x(t) = 6t^2 - 14t + 10,\ y(t) = -2t^3 + 9t^2 + 4t$:

$$\frac{dx}{dt} = 12t - 14, \quad \frac{dy}{dt} = -6t^2 + 18t + 4, \quad \frac{dy}{dx} = \frac{-6t^2 + 18t + 4}{12t - 14}.$$

Second derivative:

$$\frac{d^2y}{dx^2} = \frac{\dfrac{d}{dt}\left(\dfrac{dy}{dx}\right)}{\dfrac{dx}{dt}}.$$

11.2 Arc Length of Parametric Curves

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \, dt \qquad \text{(from } t = a \text{ to } t = b\text{).}$$

11.3 Vector-Valued Functions

If $\vec{f}(t) = \langle x(t), y(t)\rangle = x(t)\,\vec{i} + y(t)\,\vec{j}$, then

$$\vec{f}\,'(t) = \langle x'(t), y'(t)\rangle, \qquad \int \vec{f}(t)\,dt = \left\langle \int x(t)\,dt,\ \int y(t)\,dt \right\rangle.$$

11.4 Polar Coordinates and Differentiation

With $x = r\cos\theta$, $y = r\sin\theta$ and $r = f(\theta)$, the slope is

$$\frac{dy}{dx} = \frac{\dfrac{d}{d\theta}[f(\theta)\sin\theta]}{\dfrac{d}{d\theta}[f(\theta)\cos\theta]} = \frac{f'(\theta)\sin\theta + f(\theta)\cos\theta}{f'(\theta)\cos\theta - f(\theta)\sin\theta}.$$

11.5 Area in Polar Coordinates

Single curve:

$$A = \int_a^b \frac{1}{2}r^2\,d\theta.$$

Region between two curves (with $0 \le r_1(\theta) \le r_2(\theta)$, $a \le \theta \le b$):

$$A = \int_a^b \frac{1}{2}\left(r_2^2 - r_1^2\right)d\theta.$$

12. Infinite Sequences and Series

12.1 Sequences and Convergence

A sequence has limit L if for any $\varepsilon > 0$ there is a positive integer N such that $|a_n - L| < \varepsilon$ for all $n \ge N$; then the sequence **converges**, $\lim_{n\to\infty} a_n = L$. If there is no finite limit, it **diverges**.

12.2 Series and Partial Sums

A series is the sum of all terms of a sequence:

$$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + a_3 + \cdots, \qquad S_n = \sum_{k=1}^{n} a_k \ (n\text{th partial sum}).$$

If the sequence of partial sums converges to a limit S, the series converges and $S = \sum_{k=1}^{\infty} a_k$. A divergent series has no sum.

12.3 p-Series Test

$$\sum_{n=1}^{\infty} \frac{1}{n^p}: \qquad \text{converges if } p > 1, \qquad \text{diverges if } 0 < p \le 1.$$

12.4 Comparison Test

Let $0 < a_n \leq b_n$ for all n.

1. If $\sum b_n$ converges, then $\sum a_n$ converges.
2. If $\sum a_n$ diverges, then $\sum b_n$ diverges.

12.5 Limit Comparison Test

For $a_n > 0,\ b_n > 0$, if $\lim_{n\to\infty} \frac{a_n}{b_n} = L$ is finite and positive, then $\sum a_n$ and $\sum b_n$ both converge or both diverge. *Example:* $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1}$ compares with $\sum \frac{1}{n^{3/2}}$.

12.6 Ratio Test

Used for series with complicated forms. For $\sum a_n$ with nonzero terms:

1. Converges absolutely if $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$.
2. Diverges if the limit > 1 or $= \infty$.
3. Inconclusive if the limit $= 1$.

Example. $\sum_{n=0}^{\infty} \frac{n^2 2^{n+1}}{3^n}$:

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \frac{2}{3} \lim_{n\to\infty} \frac{(n+1)^2}{n^2} = \frac{2}{3} < 1 \Rightarrow \text{converges.}$$

12.7 Alternating Series Test

For $a_n > 0$, the alternating series $\sum (-1)^n a_n$ and $\sum (-1)^{n+1} a_n$ converge if

$$(1)\ \lim_{n\to\infty} a_n = 0 \qquad \text{and} \qquad (2)\ a_{n+1} \leq a_n \text{ for all } n.$$

Example: $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$, with $a_n = \frac{1}{n} \to 0$ and $a_{n+1} = \frac{1}{n+1} \leq \frac{1}{n}$.

12.8 Absolute and Conditional Convergence

- **Absolute convergence:** if $\sum |a_n|$ converges, then $\sum a_n$ converges.
- **Conditional convergence:** $\sum a_n$ converges but $\sum |a_n|$ diverges.

12.9 Radius and Interval of Convergence

Only the geometric test and ratio test can be used to determine the radius and interval of convergence. **Example.** For $\sum_{n=0}^{\infty} 3(x-2)^n$, with $r = x - 2$:

$$|x-2| < 1 \Rightarrow \text{radius} = 1, \qquad -1 < x-2 < 1 \Rightarrow 1 < x < 3 \text{ (interval).}$$

12.10 Taylor and Maclaurin Series

If f has derivatives of all orders at $x = c$, the Taylor series for f at c is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n = f(c) + f'(c)(x-c) + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n + \cdots .$$

If $c = 0$, it is the **Maclaurin series**.

12.11 Error Bound (Alternating Series)

If a convergent alternating series satisfies $a_{n+1} \leq a_n$, the remainder R_n in approximating S by S_n is less than the first neglected term:

$$|S - S_n| = |R_n| \leq a_{n+1}.$$

12.12 Taylor's Theorem

If f is differentiable through order $n+1$ on an interval I containing c, then for each x in I there exists z between x and c with

$$f(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n + R_n(x),$$

$$\text{where } R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!}(x-c)^{n+1}, \qquad c \leq z \leq x_0.$$